Izmir Institute of Technology

Electrical and Electronics Eng. Dep.

EE 471 Modern Software Development Practices and Technologies

Mini Project #2

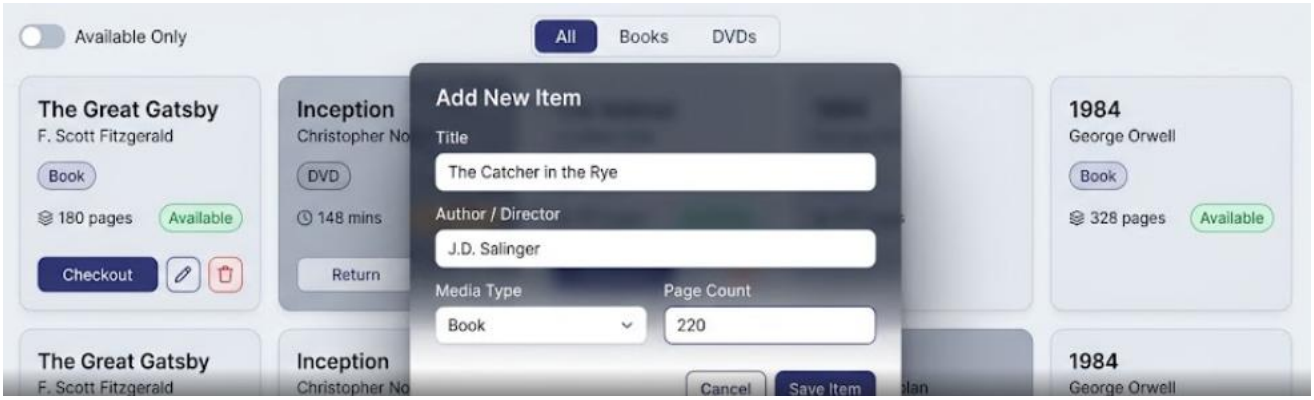
Deadline: Tuesday 03/11/2026 @23:59

Security & Environment Setup

- Store all API keys (e.g., Azure Speech key, regions) inside a local .env file.
- Ensure .env is listed in your .gitignore so secrets are never pushed to GitHub.

Voice-activated Library App

Starting point: Use the merged main branch from Week 4 for your work, *Library App*.



In week 4, you developed a library app working with the following endpoints below (Every response is JSON. Error response {"error": "<message>"}.):

Endpoint	Purpose	Success	Errors
POST /api/items	Create a Book or DVD	201 + the new item	400
GET /api/items	List items; supports ?available=true and ?type=book	200	400 bad filter
GET /api/items/<id>	Get one item	200	404
PUT /api/items/<id>	Replace all editable fields	200	400, 404
PATCH /api/items/<id>	Change some editable fields	200	400, 404
DELETE /api/items/<id>	Remove an item	204	404, 409 if checked out
POST /api/items/<id>/checkout	Check an item out	200	404, 409 already out
POST /api/items/<id>/return	Return an item	200	404, 409 not checked out

Improve your Library app with voice activation using Azure Speech Services. Each endpoint should allow for taking different voice inputs to run. At the end of each run, your app should also speak out the completed job. **Voice Interaction Examples:**

- **Create Item (POST):**
  - **User Spoken Input:** *"Create item, type book, title The Catcher in the Rye, author J.D. Salinger, pages 220"*
  - **Backend Action:** Calls POST /api/items. The backend assigns the ID (e.g., id: 3).
  - **Librarian Audio Response:** *"Successfully created book 'The Catcher in the Rye' with ID 3."*
- **Checkout Item (POST):**
  - **User Spoken Input:** *"Check out item 3"*
  - **Backend Action:** Calls POST /api/items/3/checkout.
  - **Librarian Audio Response:** *"Item 3, 'The Catcher in the Rye', has been checked out."*
- **Error Handling (409 / 404):**
  - **User Spoken Input:** *"Check out item 3"* (again)
  - **Librarian Audio Response:** *"Sorry, item 3 is already checked out."*

Deliverables:

1) Github repository link.
2) “video-demo.avi”: A screen recording video of you testing each of the endpoints with voice inputs.
3) “video-explanation.avi”: A screen recording video of you explaining the function of each code block you use in your repository.
4) “PROMPTS.md” Your vibe coding transcripts.